

Kalkulus dan Aljabar

Suprayogi

Materi

- Turunan dari suatu fungsi
- Matriks
- Pemecahan akar persamaan polynomial
- Pemecahan sistem persamaan linier

Turunan dari suatu fungsi

$f(x)$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Contoh:

$$f(x) = 2x$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{2(x + \Delta x) - 2(x)}{\Delta x}$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{2x + 2\Delta x - 2x}{\Delta x}$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} 2$$

$$\frac{df}{dx} = 2$$

Notasi-Notasi

$$\frac{df}{dx} = \frac{d}{dx} f(x) = f'(x)$$

Turunan dari fungsi konstanta

$$f(x) = c$$

$$\frac{df}{dx} = 0$$

Contoh:

$$f(x) = 2$$

$$\frac{df}{dx} = 0$$

Turunan dari fungsi pangkat

$$f(x) = x^n$$

$$\frac{df}{dx} = nx^{n-1}$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^n - x^n}{\Delta x}$$

$$(x + \Delta x)^n = x^n + nx^{n-1}\Delta x + \frac{n(n-1)}{2!}x^{n-2}(\Delta x)^2 + \frac{n(n-1)(n-2)}{3!}x^{n-3}(\Delta x)^3 + \dots$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{x^n + nx^{n-1}\Delta x + \frac{n(n-1)}{2!}x^{n-2}(\Delta x)^2 + \frac{n(n-1)(n-2)}{3!}x^{n-3}(\Delta x)^3 + \dots - x^n}{\Delta x}$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{nx^{n-1}\Delta x + \frac{n(n-1)}{2!}x^{n-2}(\Delta x)^2 + \frac{n(n-1)(n-2)}{3!}x^{n-3}(\Delta x)^3 + \dots}{\Delta x}$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}(\Delta x) + \frac{n(n-1)(n-2)}{3!}x^{n-3}(\Delta x)^2 + \dots$$

$$\frac{df}{dx} = nx^{n-1}$$

Contoh:

$$f(x) = x^3$$

$$\frac{df}{dx} = 3x^2$$

Aturan Penjumlahan

$$f(x) = g(x) \pm h(x)$$

$$\frac{df}{dx} = \frac{dg}{dx} \pm \frac{dh}{dx}$$

Contoh

$$f(x) = x^2 + (x + 2)$$

$$g(x) = x^2 \Rightarrow \frac{dg}{dx} = 2x$$

$$h(x) = x + 2 \Rightarrow \frac{dh}{dx} = 1$$

$$\frac{df}{dx} = 2x + 1$$

Aturan Perkalian

$$f(x) = g(x)h(x)$$

$$\frac{df}{dx} = \frac{dg}{dx}h(x) + g(x)\frac{dh}{dx}$$

Contoh

$$f(x) = x^2(x + 2)$$

$$g(x) = x^2 \Rightarrow \frac{dg}{dx} = 2x$$

$$h(x) = x + 2 \Rightarrow \frac{dh}{dx} = 1$$

$$\frac{df}{dx} = 2x(x + 2) + x^2(1) = 2x^2 + 4x + x^2 = 3x^2 + 4x$$

Aturan Perkalian

$$f(x) = \frac{g(x)}{h(x)}; h(x) \neq 0$$

$$\frac{df}{dx} = \frac{\frac{dg}{dx} h(x) - g(x) \frac{dh}{dx}}{(h(x))^2}$$

Contoh

$$f(x) = \frac{x^2}{x+2}$$

$$g(x) = x^2 \Rightarrow \frac{dg}{dx} = 2x$$

$$h(x) = (x+2) \Rightarrow \frac{dh}{dx} = 1$$

$$\frac{df}{dx} = \frac{2x(x+2) - x^2(1)}{(x+2)^2} = \frac{2x^2 + 4x - x^2}{x^2 + 2x + 4} = \frac{x^2 + 4x}{x^2 + 2x + 4}$$

Aturan Rantai

$$y = f(u)$$

$$u = g(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Contoh

$$y = u^2 \Rightarrow \frac{dy}{du} = 2u$$

$$u = x + 2 \Rightarrow \frac{du}{dx} = 1$$

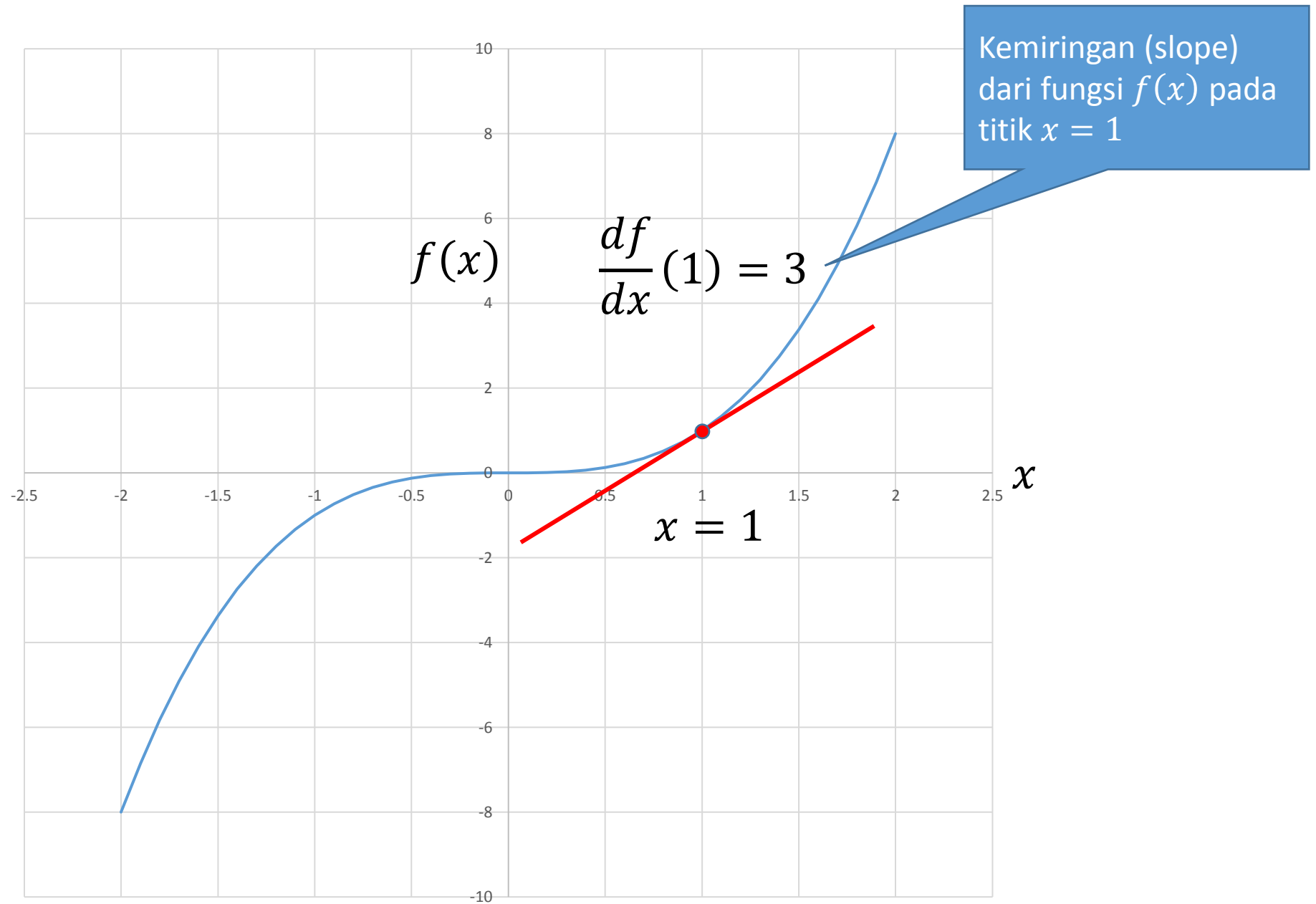
$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = 2u(1) = 2u = 2(x + 2) = 2x + 4$$

Turunan pada Suatu Titik

$$f(x) = x^3$$

$$x = 1$$

$$\left. \frac{df}{dx} \right|_{x=1} = \frac{df}{dx}(1) = \frac{d}{dx} f(1) = 3(1)^2 = 3$$



Turunan Orde Lebih Tinggi

$$f(x)$$

$$\frac{d}{dx} \left(\frac{df}{dx} \right) = \frac{d^2 f}{dx dx} = \frac{d^2 f}{dx^2}$$

Contoh:

$$f(x) = x^3$$

$$\frac{df}{dx} = 3x^2$$

$$\frac{d^2 f}{dx dx} = 6x$$

$$\frac{d^3 f}{dx dx dx} = 6$$

Turunan Parsial

$$f(x_1, x_2, \dots, x_n)$$

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x \rightarrow 0} \frac{f(x_1, \dots, x_{i-1}, x_i + \Delta x_i, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n)}{\Delta x_i}$$

Contoh

$$f(x_1, x_2) = 2x_1 + 2x_2$$

$$\frac{\partial f}{\partial x_1} = \lim_{\Delta x \rightarrow 0} \frac{(2(x_1 + \Delta x_1) + 2x_2) - (2x_1 + 2x_2)}{\Delta x_1}$$

$$\frac{\partial f}{\partial x_1} = \lim_{\Delta x \rightarrow 0} \frac{(2x_1 + 2\Delta x_1 + 2x_2) - (2x_1 + 2x_2)}{\Delta x_1}$$

$$\frac{\partial f}{\partial x_1} = \lim_{\Delta x \rightarrow 0} 2$$

$$\frac{\partial f}{\partial x_1} = 2$$

Contoh

$$f(x_1, x_2) = x_1^3 + 2x_2^2 + x_1^2x_2$$

$$\frac{\partial f}{\partial x_1} = 3x_1^2 + 2x_1x_2$$

$$\frac{\partial f}{\partial x_2} = 4x_2 + x_1^2$$

Turunan Parsial Orde Lebih Tinggi

$$f(x_1, x_2, \dots, x_n)$$

$$\frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_i \partial x_i} = \frac{\partial^2 f}{\partial x_i^2}$$

$$\left. \begin{aligned} \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) &= \frac{\partial^2 f}{\partial x_i \partial x_j} \\ \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) &= \frac{\partial^2 f}{\partial x_j \partial x_i} \end{aligned} \right\} \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

Tabel Turunan

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	nx^{n-1}	e^x	e^x
$\ln(x)$	$1/x$	$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$	$\tan(x)$	$\sec^2(x)$
$\cot(x)$	$-\operatorname{cosec}^2(x)$	$\sec(x)$	$\sec(x) \tan(x)$
$\operatorname{cosec}(x)$	$-\operatorname{cosec}(x) \cot(x)$	$\tan^{-1}(x)$	$1/(1+x^2)$
$\sin^{-1}(x)$	$1/\sqrt{1-x^2}$ for $ x < 1$	$\cos^{-1}(x)$	$-1/\sqrt{1-x^2}$ for $ x < 1$
$\sinh(x)$	$\cosh(x)$	$\cosh(x)$	$\sinh(x)$
$\tanh(x)$	$\operatorname{sech}^2(x)$	$\coth(x)$	$-\operatorname{cosech}^2(x)$
$\operatorname{sech}(x)$	$-\operatorname{sech}(x) \tanh(x)$	$\operatorname{cosech}(x)$	$-\operatorname{cosech}(x) \coth(x)$
$\sinh^{-1}(x)$	$1/\sqrt{x^2+1}$	$\cosh^{-1}(x)$	$1/\sqrt{x^2-1}$ for $x > 1$
$\tanh^{-1}(x)$	$1/(1-x^2)$ for $ x < 1$	$\coth^{-1}(x)$	$-1/(x^2-1)$ for $ x > 1$

Matriks

Matriks $m \times n$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Transpos dari Matriks

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad \text{Ukuran } m \times n$$

$$\mathbf{A}^T = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad \text{Ukuran } n \times m$$

Contoh

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \end{pmatrix}$$

$$\mathbf{A}^T = \begin{pmatrix} 1 & -2 \\ -2 & 0 \\ -1 & 2 \end{pmatrix}$$

Operasi Penjumlahan/Pengurangan

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Ukuran $m \times n$

$$\mathbf{B} = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix}$$

Ukuran $m \times n$

$$\mathbf{A} \pm \mathbf{B} = \begin{pmatrix} a_{11} \pm b_{11} & a_{12} \pm b_{12} & \cdots & a_{1n} \pm b_{1n} \\ a_{21} \pm b_{21} & a_{22} \pm b_{22} & \cdots & a_{2n} \pm b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} \pm b_{m1} & a_{m2} \pm b_{m2} & \cdots & a_{mn} \pm b_{mn} \end{pmatrix}$$

Ukuran $m \times n$

Operasi Perkalian dengan Skalar

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad \text{Ukuran } m \times n$$

$$k\mathbf{A} = \begin{pmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ ka_{21} & ka_{22} & \cdots & ka_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ka_{m1} & ka_{m2} & \cdots & ka_{mn} \end{pmatrix} \quad \text{Ukuran } m \times n$$

Operasi Perkalian antara Dua Matriks

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{pmatrix}$$

Ukuran $m \times k$

$$\mathbf{B} = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{k1} & b_{k2} & \cdots & b_{kn} \end{pmatrix}$$

Ukuran $k \times n$

$$\mathbf{AB} = \begin{pmatrix} \sum_{p=1}^k a_{1p}b_{p1} & \sum_{p=1}^k a_{1p}b_{p2} & \cdots & \sum_{p=1}^k a_{1p}b_{pn} \\ \sum_{p=1}^k a_{2p}b_{p1} & \sum_{p=1}^k a_{2p}b_{p2} & \cdots & \sum_{p=1}^k a_{2p}b_{pn} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{p=1}^k a_{mp}b_{p1} & \sum_{p=1}^k a_{mp}b_{p2} & \cdots & \sum_{p=1}^k a_{mp}b_{pn} \end{pmatrix}$$

Ukuran $m \times n$

Matriks Bujur Sangkar

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

$$m = n$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

Matriks Identitas

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ a_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

Ukuran $n \times n$

Elemen pada diagonal = 1

Elemen yang lain = 0

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Determinan dari Matriks (1 x 1)

$$\mathbf{A} = (a_{11})$$

$$\det \mathbf{A} = a_{11}$$

Determinan dari Matriks (2 x 2)

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$\det(\mathbf{A}) = |\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}$$

Contoh:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$$

$$\det(\mathbf{A}) = |\mathbf{A}| = (1)(0) - (-2)(-1) = -2$$

Determinan dari Matriks (3 x 3)

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\det(\mathbf{A}) = |\mathbf{A}| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$$

Contoh

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix}$$

$$\begin{aligned} \det(\mathbf{A}) = |\mathbf{A}| &= (1)(0)(-2) + (-2)(2)(1) + (-1)(-1)(-1) \\ &\quad - (-1)(0)(1) - (1)(2)(-1) - (-2)(-1)(-2) = 1 \end{aligned}$$

Minor

Minor pada baris i dan kolom j , $\mathbf{M}_{ij}$, adalah submatriks dari matriks $\mathbf{A}$ yang diperoleh dengan menghapus baris i dan kolom j

Contoh

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix}$$

$$\mathbf{M}_{11} = \begin{pmatrix} 0 & 2 \\ -1 & -2 \end{pmatrix} \quad \mathbf{M}_{12} = \begin{pmatrix} 1 & 2 \\ -1 & -2 \end{pmatrix} \quad \mathbf{M}_{13} = \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}$$

$$\mathbf{M}_{21} = \begin{pmatrix} -2 & -1 \\ -1 & -2 \end{pmatrix} \quad \mathbf{M}_{22} = \begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix} \quad \mathbf{M}_{23} = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$$

$$\mathbf{M}_{31} = \begin{pmatrix} -2 & -1 \\ 0 & 2 \end{pmatrix} \quad \mathbf{M}_{32} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \quad \mathbf{M}_{33} = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$$

Kofaktor

Misal $\mathbf{A}$ adalah matriks bujursangkar ukuran $n \times n$

Cofactor dari $\mathbf{A}$ adalah matriks yang elemen i dan elemen j adalah

$$c_{ij} = (-1)^{i+j} \det(\mathbf{M}_{ij})$$

Contoh

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix}$$

Cofactor dari $\mathbf{A}$

$$\mathbf{C} = \begin{pmatrix} 2 & 0 & 1 \\ -3 & -1 & -1 \\ -4 & -1 & -2 \end{pmatrix}$$

$$\mathbf{M}_{11} = \begin{pmatrix} 0 & 2 \\ -1 & -2 \end{pmatrix}$$

$$\det(\mathbf{M}_{11}) = 2$$

$$c_{11} = (-1)^{1+1}(2) = 2$$

$$\mathbf{M}_{12} = \begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}$$

$$\det(\mathbf{M}_{12}) = 0$$

$$c_{12} = (-1)^{1+2}(0) = 0$$

$$\mathbf{M}_{13} = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix}$$

$$\det(\mathbf{M}_{13}) = 1$$

$$c_{13} = (-1)^{1+3}(-1) = 1$$

$$\mathbf{M}_{21} = \begin{pmatrix} -2 & -1 \\ -1 & -2 \end{pmatrix}$$

$$\det(\mathbf{M}_{21}) = 3$$

$$c_{21} = (-1)^{2+1}(3) = -3$$

$$\mathbf{M}_{22} = \begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix}$$

$$\det(\mathbf{M}_{22}) = -1$$

$$c_{22} = (-1)^{2+2}(-1) = -1$$

$$\mathbf{M}_{23} = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$$

$$\det(\mathbf{M}_{23}) = 1$$

$$c_{23} = (-1)^{2+3}(1) = -1$$

$$\mathbf{M}_{31} = \begin{pmatrix} -2 & -1 \\ 0 & 2 \end{pmatrix}$$

$$\det(\mathbf{M}_{31}) = -4$$

$$c_{31} = (-1)^{3+1}(-4) = -4$$

$$\mathbf{M}_{32} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

$$\det(\mathbf{M}_{32}) = 1$$

$$c_{32} = (-1)^{3+2}(1) = -1$$

$$\mathbf{M}_{33} = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$$

$$\det(\mathbf{M}_{33}) = -2$$

$$c_{33} = (-1)^{3+3}(-2) = -2$$

Determinan dari Matriks ($n \times n$)

$$\det(\mathbf{A}) = \sum_{j=1}^n a_{ij} (-1)^{i+j} \det(\mathbf{M}_{ij}); \text{ untuk } i \text{ tertentu}$$

$$\det(\mathbf{A}) = \sum_{j=1}^n a_{ij} c_{ij}; \text{ untuk } i \text{ tertentu}$$

$$\det(\mathbf{A}) = \sum_{i=1}^n a_{ij} (-1)^{i+j} \det(\mathbf{M}_{ij}); \text{ untuk } j \text{ tertentu}$$

$$\det(\mathbf{A}) = \sum_{i=1}^n a_{ij} c_{ij}; \text{ untuk } j \text{ tertentu}$$

Contoh: $\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix}$ $\mathbf{C} = \begin{pmatrix} 2 & 0 & 1 \\ -3 & -1 & -1 \\ -4 & -1 & -2 \end{pmatrix}$

$$\det(\mathbf{A}) = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} = (1)(2) + (-2)(0) + (-1)(1) = 1$$

$$\det(\mathbf{A}) = a_{11}c_{11} + a_{21}c_{21} + a_{31}c_{31} = (1)(2) + (-1)(-3) + (1)(-4) = 1$$

Invers dari Matriks

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$\mathbf{A}^{-1} = \frac{\text{Transpos dari Cofactor dari } \mathbf{A}}{\det(\mathbf{A})} = \frac{\mathbf{C}^T}{\det(\mathbf{A})}$$

Contoh:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix} \quad \det(\mathbf{A}) = 1$$

$$\mathbf{C} = \begin{pmatrix} 2 & 0 & 1 \\ -3 & -1 & -1 \\ -4 & -1 & -2 \end{pmatrix} \quad \mathbf{C}^T = \begin{pmatrix} 2 & -3 & -4 \\ 0 & -1 & -1 \\ 1 & -1 & -2 \end{pmatrix}$$

$$\mathbf{A}^{-1} = \frac{\begin{pmatrix} 2 & -3 & -4 \\ 0 & -1 & -1 \\ 1 & -1 & -2 \end{pmatrix}}{1} = \begin{pmatrix} 2 & -3 & -4 \\ 0 & -1 & -1 \\ 1 & -1 & -2 \end{pmatrix}$$

Invers dari Matriks 2 x 2

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$$

$$\mathbf{A}^{-1} = \frac{1}{(-1)(0) - (2)(1)} \begin{pmatrix} 0 & -(-2) \\ -(-1) & -1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -0,5 & -0,5 \end{pmatrix}$$

Menggunakan Operasi Baris Elementer

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad \mathbf{I} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ a_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

$$\mathbf{A}' = (\mathbf{A}|\mathbf{I})$$



Operasi baris elementer

$$\mathbf{A}' = (\mathbf{I}|\mathbf{A}^{-1})$$

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix}$$

$$\mathbf{A}' = \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 & 1 & 0 \\ 1 & -1 & -2 & 0 & 0 & 1 \end{array} \right)$$

$$\mathbf{A}' = \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array} \right)$$

$$\mathbf{A}' = \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & -1 & 0 \\ 0 & 1 & -0.5 & -0.5 & -0.5 & 0 \\ 0 & 0 & -0.5 & -0.5 & 0.5 & 1 \end{array} \right)$$

$$\mathbf{A}^{-1} = \begin{pmatrix} 2 & -3 & -4 \\ 0 & -1 & -1 \\ 1 & -1 & -2 \end{pmatrix}$$

$$\mathbf{A}' = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -3 & -4 \\ 0 & 1 & 0 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 & -1 & -2 \end{array} \right)$$

Matriks Singular

Matriks singular $\rightarrow$ determinan = 0 $\rightarrow$ tidak ada invers

$$AA^{-1} = I$$

Matriks A $\rightarrow$ matriks tak singular $\rightarrow$ determinan $\neq 0$

Principal Minor

Misal $\mathbf{A}$ adalah matriks bujursangkar ukuran $n \times n$

Principal Minor orde k adalah submatriks
dari matriks $\mathbf{A}$ yang diperoleh dengan menghapus
baris $n - k$ dan kolom $n - k$

Matriks bujursangkar $\mathbf{A}$ ukuran $n \times n \rightarrow 2^n - 1$ principal minor

Contoh

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix}$$

Principal Minor orde 1

$$(-2) \quad (0) \quad (1)$$

Principal Minor orde 2

$$\begin{pmatrix} 0 & -2 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$$

Principal Minor orde 3

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix}$$

Leading Principal Minor

Misal $\mathbf{A}$ adalah matriks bujursangkar ukuran $n \times n$

Leading Principal Minor orde k adalah submatriks dari matriks $\mathbf{A}$ yang diperoleh dengan menghapus baris $n - k$ terakhir dan kolom $n - k$ terakhir

Matriks bujursangkar $\mathbf{A}$ ukuran $n \times n \rightarrow n$ leading principal minor

Contoh

Matriks **A** ukuran 3×3

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$$

Leading Principal Minor orde 1 (1)

Leading Principal Minor orde 2 $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$

Leading Principal Minor orde 3 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$

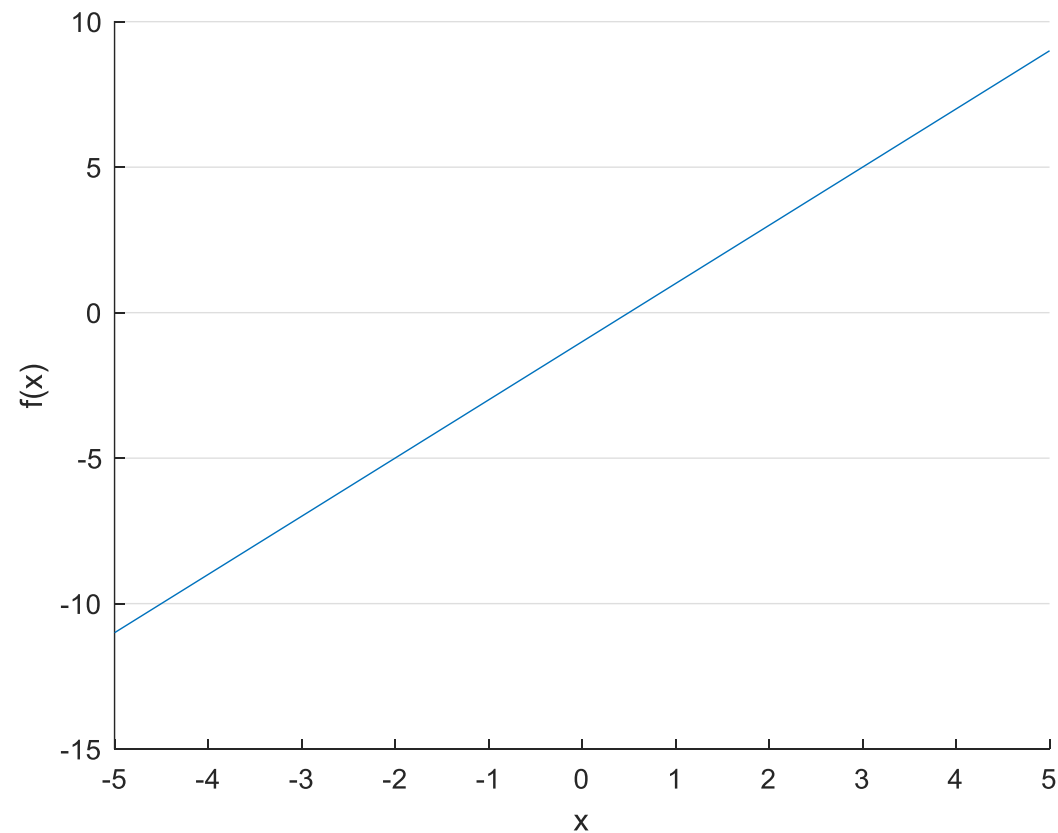
Penentuan Akar Persamaan

Fungsi Polinomial orde n

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 x^0$$

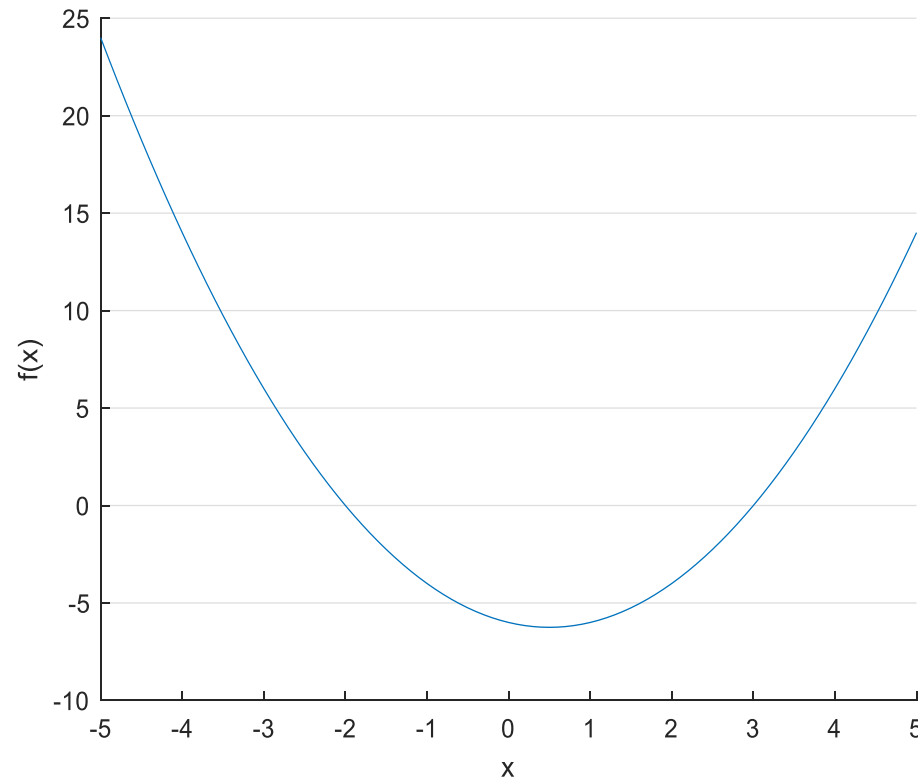
Contoh:

$$f(x) = 2x - 1$$



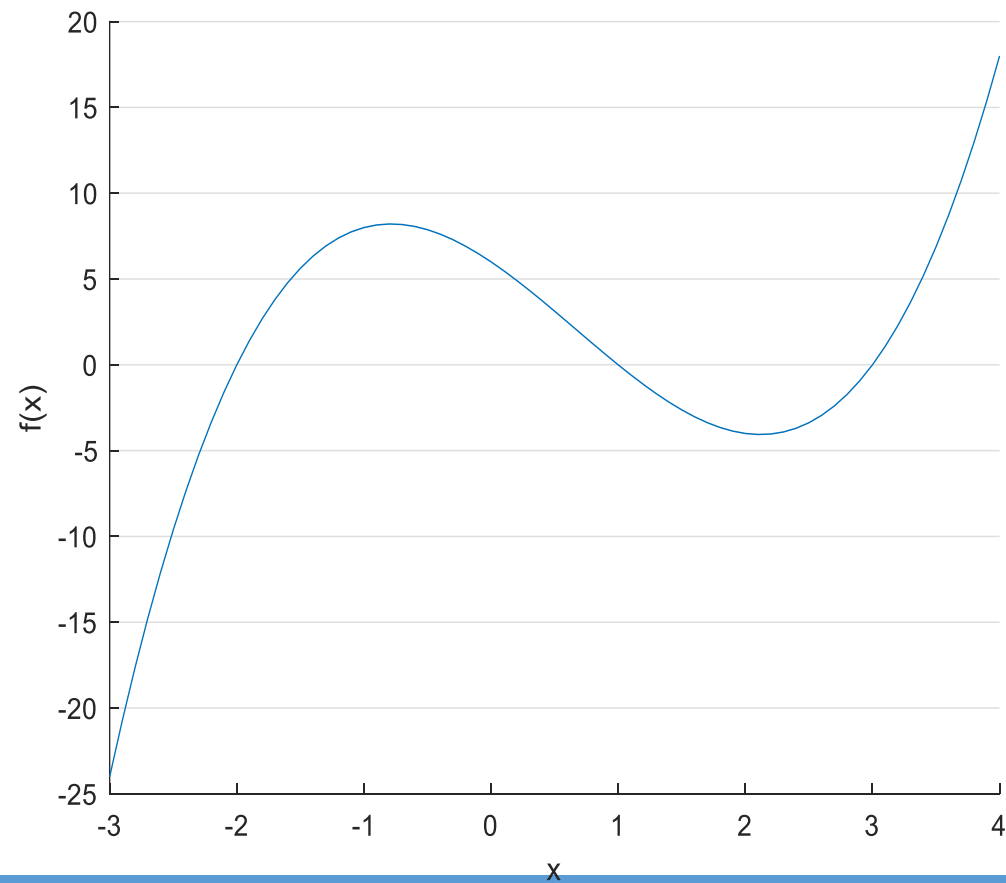
Contoh:

$$f(x) = x^2 - x - 6$$



Contoh:

$$f(x) = x^3 - 2x^2 - 5x + 6$$



Akar dari Persamaan Polinomial

Misal $f(x)$ merupakan fungsi polinomial derajat n .

$x = r$ adalah akar dari $f(x)$ jika $x = r$ memenuhi
$$f(x) = 0$$

Jika $f(x)$ merupakan fungsi polinomial derajat n , maka
 $f(x)$ akan memiliki n akar yang beberapa diantaranya
dapat berulang (sama)

Misal $f(x)$ merupakan fungsi polinomial derajat n .

Jika r adalah akar dari $f(x)$, maka $(x - r)$ adalah faktor dari $f(x)$

Jika $(x - r)$ adalah faktor dari $f(x)$, maka r adalah akar dari $f(x)$

Faktor

$$f(x) = x^2 - x - 6$$

Faktor

$$f(x) = (x - 3)(x + 2)$$

Akar

$$x = 3$$

$$x = -2$$

Akar Persamaan Polinomial Orde 1

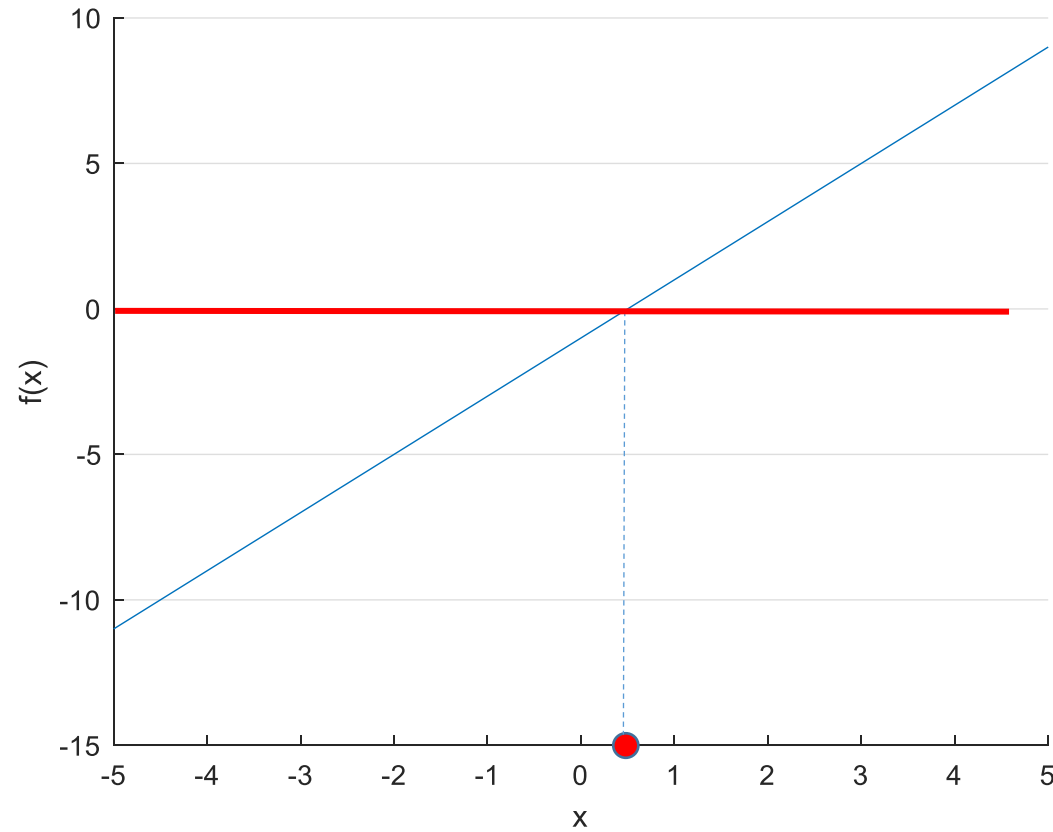
$$f(x) = a_1x + a_0$$

$$x = -\frac{a_0}{a_1}$$

Contoh

$$f(x) = 2x - 1 = 0$$

$$x = \frac{1}{2}$$



Akar Persamaan Polinomial Orde 2

$$f(x) = a_2x^2 + a_1x + a_0 = 0$$

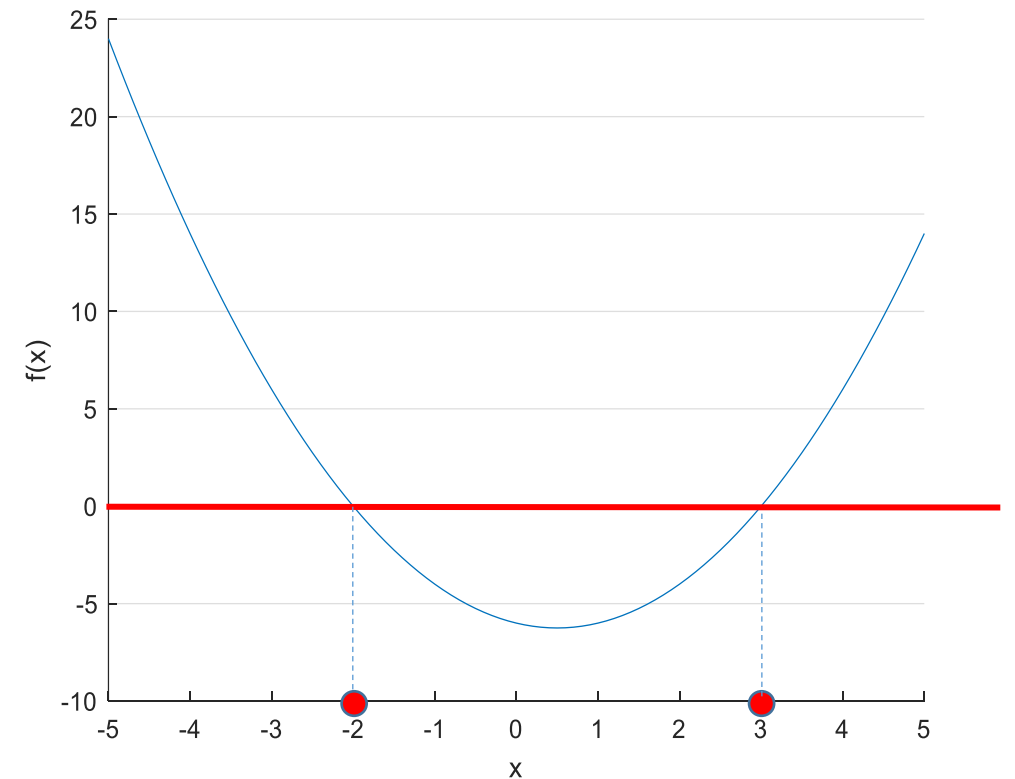
$$x = \frac{-a_1 \pm \sqrt{(a_1)^2 - 4a_2a_0}}{2a_2}$$

Contoh

$$f(x) = x^2 - x - 6 = 0$$

$$x = \frac{-(-1) + \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)} = \frac{1 + \sqrt{25}}{2} = 3$$

$$x = \frac{-(-1) - \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)} = \frac{1 - \sqrt{25}}{2} = -2$$



Teorema

Jika $f(x)$ merupakan fungsi polinomial derajat n dan $x = r$ adalah, maka $f(x)$ dapat dinyatakan dengan

$$f(x) = (x - r)q(x)$$

dengan $q(x)$ merupakan fungsi polinomial derajat $n - 1$ yang dapat diperoleh dengan membagi $f(x)$ dengan $(x - r)$

Teorema Akar Rasional

Misal $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$
dengan koefisien $a_n, a_{n-1}, \dots, a_0$ adalah bilangan bulat

Misal p adalah faktor a_0

Misal q adalah faktor a_n

Bilangan rasional

$$x = \frac{p}{q}$$

adalah akar dari $f(x)$

Contoh:

$$f(x) = x^3 - 2x^2 - 5x + 6 = 0$$

$$a_0 = 6 \rightarrow p = \pm 1, \pm 2, \pm 3, \pm 6$$

$$a_3 = 1 \rightarrow q = \pm 1$$

$$\text{Fraksi yang mungkin} \rightarrow \frac{\pm 1}{\pm 1}, \frac{\pm 2}{\pm 1}, \frac{\pm 3}{\pm 1}, \frac{\pm 6}{\pm 1}$$

$$\pm 1, \pm 2, \pm 3, \pm 6$$

Misal akar $x = 1 \rightarrow$ faktor $(x - 1)$

Bagi $f(x)$ dengan $(x - 1)$

	x^2	$-x$	-6	
$x - 1$	x^3	$-2x^2$	$-5x$	$+ 6$
	x^3	$-x^2$		$-$
	0	$-x^2$	$-5x$	$+ 6$
	0	$-x^2$	$+ x$	$-$
	0	0	$-6x$	$+ 6$
	0	0	$-6x$	$+ 6$
	0	0	0	0

Sisa pembagian = 0 $\rightarrow (x - 1)$ adalah faktor dari $f(x)$

$$f(x) = (x - 1)(x^2 - x - 6) = 0$$

$$f(x) = (x - 1)(x^2 - x - 6) = 0$$

Akar dapat diperoleh
dengan pemecahan
persamaan polinomial
orde 2

$$x = \frac{-(-1) + \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)} = \frac{1 + \sqrt{25}}{2} = \frac{1 + 5}{2} = 3$$

$$x = \frac{-(-1) - \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)} = \frac{1 - \sqrt{25}}{2} = \frac{1 - 5}{2} = -2$$

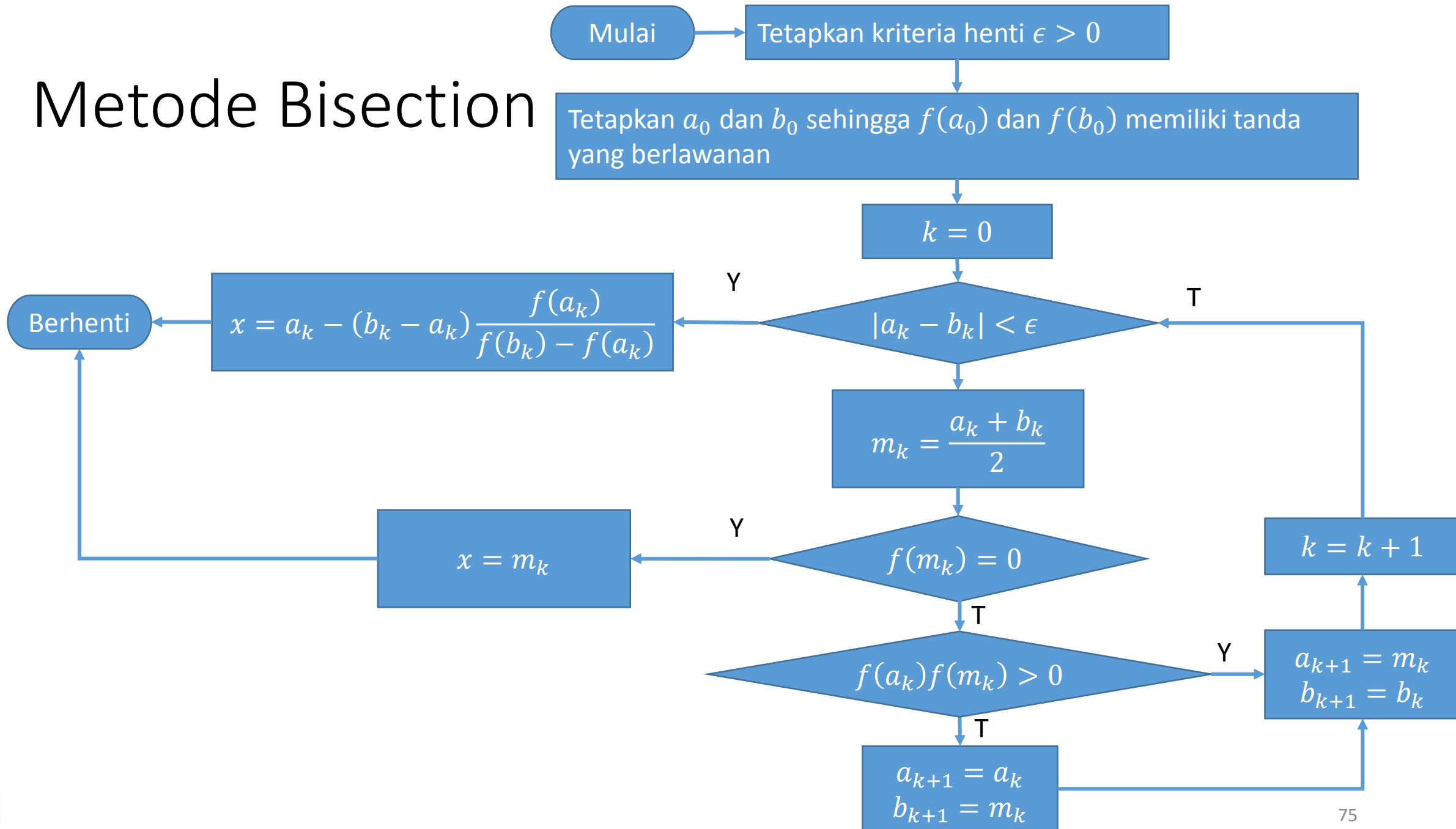
$$f(x) = (x - 1)(x - 3)(x + 2) = 0$$

Akar-akar: $x = 1, x = 3, x = -2$

Metode Numerik untuk Mencari Akar

- Metode Bisection
- Metode Newton-Raphson

Metode Bisection



$$f(x) = x^2 - x - 6$$

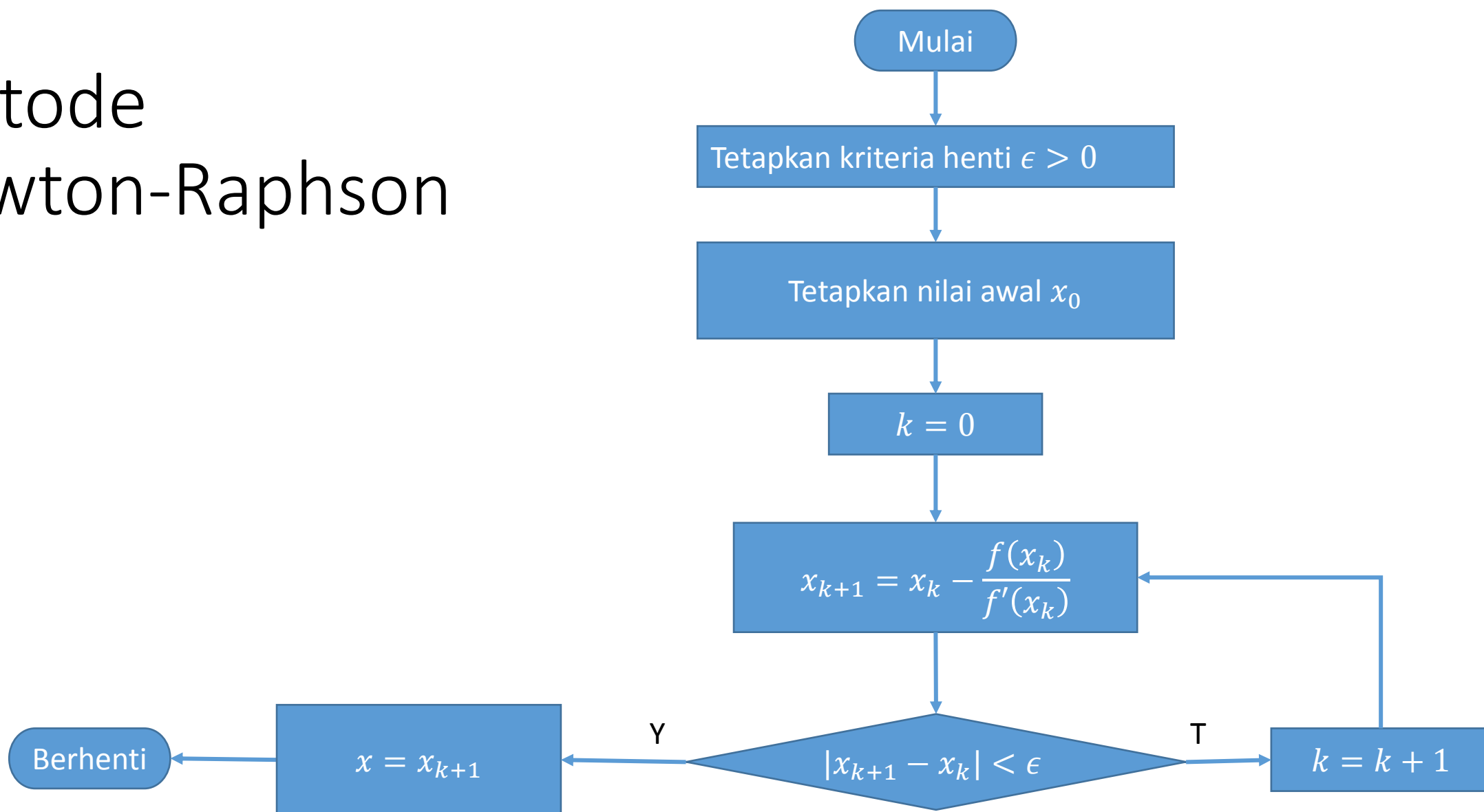
Tentukan akar dari $f(x)$

$$a_0 = 1 \quad b_0 = 4 \quad \epsilon = 0,001$$

k	a_k	b_k	$ a_k - b_k $	$f(a)$	$f(b)$	$f(m)$	$f(m)$	$f(a)f(m)$	x
0	1.000000	4.000000	3.000000	-6.000000	6.000000	2.500000	-2.250000	13.500000	2.5000
1	2.500000	4.000000	1.500000	-2.250000	6.000000	3.250000	1.312500	-2.953125	2.9091
2	2.500000	3.250000	0.750000	-2.250000	1.312500	2.875000	-0.609375	1.371094	2.9737
3	2.875000	3.250000	0.375000	-0.609375	1.312500	3.062500	0.316406	-0.192810	2.9939
4	2.875000	3.062500	0.187500	-0.609375	0.316406	2.968750	-0.155273	0.094620	2.9984
5	2.968750	3.062500	0.093750	-0.155273	0.316406	3.015625	0.078369	-0.012169	2.9996
6	2.968750	3.015625	0.046875	-0.155273	0.078369	2.992188	-0.039001	0.006056	2.9999
7	2.992188	3.015625	0.023438	-0.039001	0.078369	3.003906	0.019547	-0.000762	3.0000
8	2.992188	3.003906	0.011719	-0.039001	0.019547	2.998047	-0.009762	0.000381	3.0000
9	2.998047	3.003906	0.005859	-0.009762	0.019547	3.000977	0.004884	-0.000048	3.0000
10	2.998047	3.000977	0.002930	-0.009762	0.004884	2.999512	-0.002441	0.000024	3.0000
11	2.999512	3.000977	0.001465	-0.002441	0.004884	3.000244	0.001221	-0.000003	3.0000
12	2.999512	3.000244	0.000732	-0.002441	0.001221	2.999878	-0.000610	0.000001	3.0000

$$x = 3.000000$$

Metode Newton-Raphson



$$f(x) = x^2 - x - 6$$

Tentukan akar dari $f(x)$, yaitu nilai x sehingga $f(x) = 0$

$$f'(x) = 2x - 1$$

$$x_0 = 4 \quad \epsilon = 0,001$$

k	x	$ x_{k+1} - x_k $	$f(x)$	$f'(x)$
0	4.000000		6.000000	7.000000
1	3.142857	0.857143	0.734694	5.285714
2	3.003861	0.138996	0.019320	5.007722
3	3.000003	0.003858	0.000015	5.000006
4	3.000000	0.000003		

$$x = 3.000000$$

$$f(x) = x^2 - x - 6 = 0$$

$$f'(x) = 2x - 1$$

k	x	$ x_{k+1} - x_k $	$f(x)$	$f'(x)$
0	4.000000		6.000000	7.000000
1	3.142857	0.857143	0.734694	5.285714
2	3.003861	0.138996	0.019320	5.007722
3	3.000003	0.003858	0.000015	5.000006
4	3.000000	0.000003		

$$x = 3.000000$$

Pemecahan Sistem Persamaan Linier

Sistem Persamaan Linier

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

Sistem Persamaan Linier – Notasi Matriks

$$\begin{pmatrix} a_{11} & a_{11} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

$$\mathbf{Ax} = \mathbf{b}$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{11} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Pemecahan dengan Operasi Matriks

$$\mathbf{Ax} = \mathbf{b}$$



$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

Contoh:

$$x_1 - 2x_2 - x_3 = -2$$

$$-x_1 \quad \quad + 2x_3 = -6$$

$$x_1 - x_2 - 2x_3 = 4$$

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} -2 \\ -6 \\ 4 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_n \end{pmatrix}$$

$$\mathbf{A}^{-1} = \begin{pmatrix} 2 & -3 & -4 \\ 0 & -1 & -1 \\ 1 & -1 & -2 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_n \end{pmatrix} = \begin{pmatrix} 2 & -3 & -4 \\ 0 & -1 & -1 \\ 1 & -1 & -2 \end{pmatrix} \begin{pmatrix} -2 \\ -6 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ -4 \end{pmatrix}$$

Eliminasi Gauss-Jordan

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$\vdots$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$



Operasi baris elementer

$$x_1 + 0x_2 + \cdots + 0x_n = b'_1$$

$$0x_1 + x_2 + \cdots + 0x_n = b'_2$$

$\vdots$

$$0x_1 + 0x_2 + \cdots + x_n = b'_n$$

Contoh

$$x_1 - 2x_2 - x_3 = -2$$

$$-x_1 + 2x_3 = -6$$

$$x_1 - x_2 - 2x_3 = 4$$

$$\mathbf{x} = \begin{pmatrix} -2 \\ 2 \\ -4 \end{pmatrix}$$

x_1	x_2	x_3	b
1	-2	-1	-2
-1	0	2	-6
1	-1	-2	4

x_1	x_2	x_3	b
1	-2	-1	-2
0	-2	1	-8
0	1	-1	6

x_1	x_2	x_3	b
1	0	-2	6
0	1	-0.5	4
0	0	-0.5	2

x_1	x_2	x_3	b
1	0	0	-2
0	1	0	2
0	0	1	-4

Selesai